

# Introduction to R

Advanced Statistics Lab

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# Get R

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From CRAN

RStudio

Ubuntu:

```
sudo apt install r-base
```

# Some Introduction Material

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From the developers: [Introduction to R](#)

RStudio: [Documentation](#)

[R for Data Science](#)

# Background

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- R is a general programming language with a specialization for statistics.
- R is generally used through a REPL and interpreted live.
- Functional programming paradigm: all about objects and functions.
- Major appeal: packages, graphics, and scientific workflows.
- Major drawback: scripts can grow without structure.

# Assigning And Manipulating Variables

Note: R has two assignment operators, `<-` and `=`.

```
x = 5  
print(x)
```

```
[1] 5
```

```
x
```

```
[1] 5
```

```
y = x  
print(y)
```

```
[1] 5
```

```
x = 3  
  
print(x)
```

```
[1] 3
```

```
print(y)
```

```
[1] 5
```

# Working With Data Frames

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)
```

```
str(dat)
```

```
'data.frame':   3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : num  22 43 54
```

```
names = dat$name  
ages = dat[, "age"]
```

```
str(names)
```

```
chr [1:3] "Adam" "Paul" "Elly"
```

```
str(ages)
```

```
num [1:3] 22 43 54
```

# Subsetting Data Frames

```
dat[dat$name == "Paul", ]
```

```
  name age  
2 Paul  43
```

```
dat[dat$age %in% c(22, 54), ]
```

```
  name age  
1 Adam  22  
3 Elly  54
```

```
dat$id = 1:nrow(dat)  
dat[, "id_2"] = dat$id + 2
```

```
print(dat)
```

```
  name age id id_2  
1 Adam  22  1    3  
2 Paul  43  2    4  
3 Elly  54  3    5
```

```
dat$id[dat$name == "Adam"] <- 12  
print(dat)
```

```
  name age id id_2  
1 Adam  22 12    3
```

# Reading And Writing Data

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)  
  
write.table(dat, file = "my_data.csv", row.names = FALSE, sep = ",")  
  
dat_read = read.table(file = "my_data.csv", sep = ",", header = TRUE)  
str(dat_read)
```

```
'data.frame':   3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : int  22 43 54
```



# Vectors And Matrices

```
x = as.integer(1)
str(x)
```

```
int 1
```

```
is.vector(x)
```

```
[1] TRUE
```

```
y = rnorm(10)
str(y)
```

```
num [1:10] -0.6588 -0.2889 0.0847 1.4714 0.0139 ...
```

```
D = matrix(data = rnorm(5 * 5), nrow = 5, ncol = 5)
D
```

|      | [,1]       | [,2]        | [,3]        | [,4]        | [,5]       |
|------|------------|-------------|-------------|-------------|------------|
| [1,] | -0.6340910 | 0.92068999  | 0.20202515  | -0.89231290 | 1.1737426  |
| [2,] | 0.8626869  | -1.27317327 | 0.01580746  | 0.88362306  | -0.5524931 |
| [3,] | 1.7873326  | 2.10748999  | 1.33039736  | -1.70807868 | -0.2631091 |
| [4,] | -0.9274102 | 0.63603345  | -1.46394140 | 0.03020573  | 0.4391256  |
| [5,] | 0.6330386  | -0.05443828 | 1.53446063  | 0.16462659  | 0.7232650  |

```
class(D)
```

```
[1] "matrix" "array"
```

# Matrix Helpers

```
means = colMeans(D)
means
```

```
[1]  0.3443114  0.4673204  0.3237498 -0.3043872  0.3041062
```

```
d = diag(D)
d
```

```
[1] -0.63409104 -1.27317327  1.33039736  0.03020573  0.72326496
```

# Data Exploration And Visualization

```
library(dlookr)
library(ggplot2)
```

```
data(iris)
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1 ...
1 1 ...
```

```
dlookr::diagnose(iris)
```

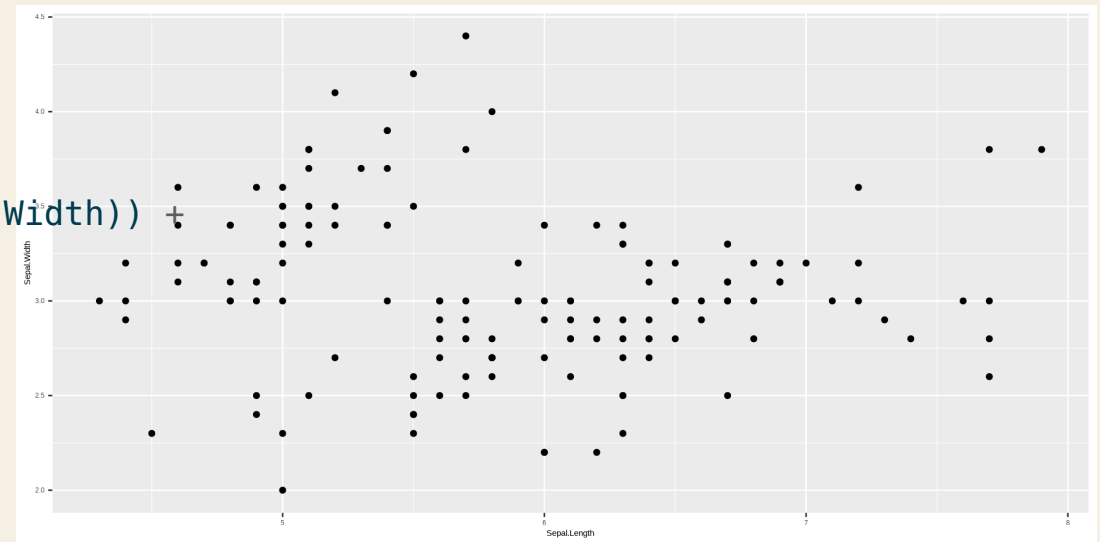
# A tibble: 5 × 6

|   | variables    | types   | missing_count | missing_percent | unique_count | unique_rate |
|---|--------------|---------|---------------|-----------------|--------------|-------------|
|   | <chr>        | <chr>   | <int>         | <dbl>           | <int>        | <dbl>       |
| 1 | Sepal.Length | numeric | 0             | 0               | 35           | 0.233       |
| 2 | Sepal.Width  | numeric | 0             | 0               | 23           | 0.153       |
| 3 | Petal.Length | numeric | 0             | 0               | 43           | 0.287       |
| 4 | Petal.Width  | numeric | 0             | 0               | 22           | 0.147       |
| 5 | Species      | factor  | 0             | 0               | 3            | 0.02        |

# Scatterplot

```
p = ggplot(iris, aes(x = Sepal.Length, y = Sepal.Width))  
  geom_point()
```

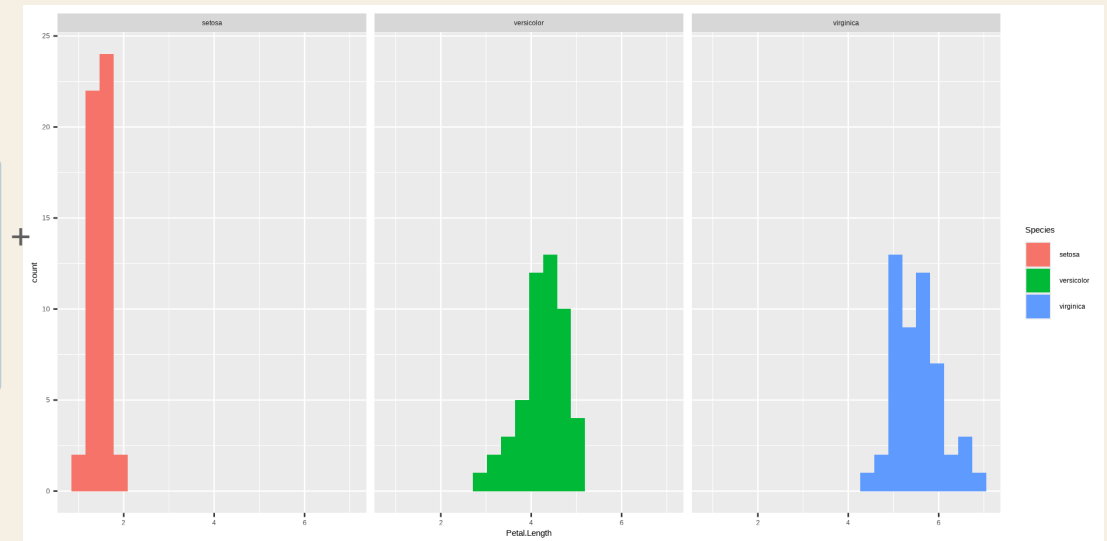
p



# Histogram

```
p2 = ggplot(iris, aes(x = Petal.Length)) +  
  geom_histogram(aes(fill = Species), bins = 20)  
  facet_wrap(~ Species)
```

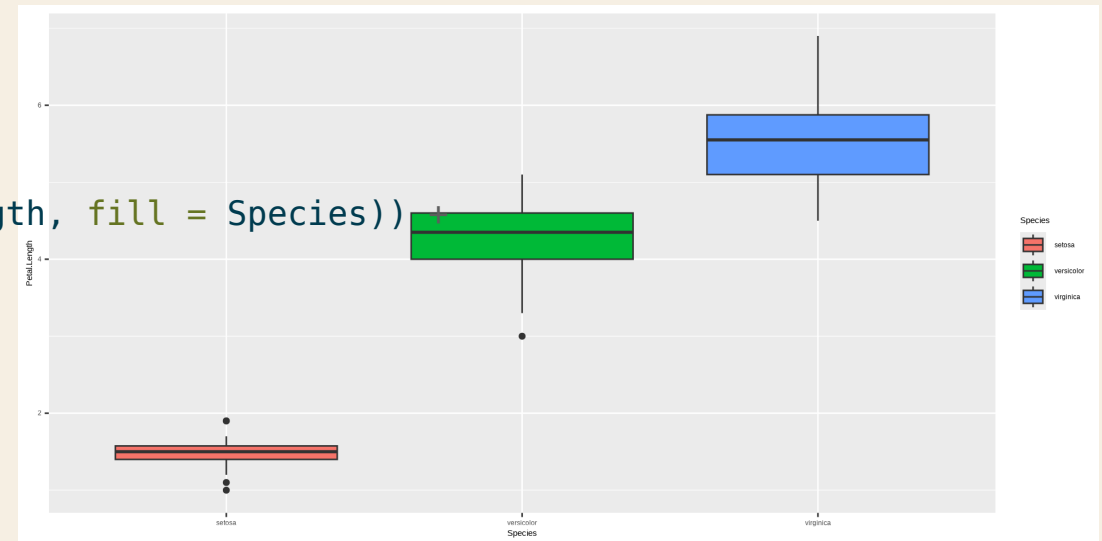
p2



# Boxplot

```
p3 = ggplot(iris, aes(x = Species, y = Petal.Length, fill = Species))  
  geom_boxplot()
```

p3



# Writing Functions

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All operators in R are functions.

```
?sum  
?`+`  
fix(`+`)
```

# A Mean Function

```
my_mean_function = function(y) {  
  n = length(y)  
  sum = 0  
  for (i in 1:n) sum = sum + y[i]  
  
  mean = sum / n  
  
  return(mean)  
}
```



# Use The Function

```
x = rnorm(1000, mean = 23.42, sd = 4)
```

```
our_mean = my_mean_function(x)  
our_mean
```

```
[1] 23.49462
```

```
mean(x)
```

```
[1] 23.49462
```